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Assignment No: 7

Problem Statement:

Implement a basic multimodal system for image captioning using CLIP.

Objective:

1. To understand multimodal AI systems that process both images and text.
2. To study the CLIP (Contrastive Language–Image Pretraining) model developed by OpenAI.
3. To implement an image caption selection system using CLIP.
4. To learn how to preprocess images and text for transformer-based models.
5. To evaluate model predictions by selecting the most relevant caption for an image.

S/W Packages and H/W apparatus used:

1. Operating System: Windows/Linux/MacOS
2. Kernel: Python 3.x
3. Tools: Jupyter Notebook / Anaconda
4. Hardware: CPU/GPU with minimum 4GB RAM
5. Libraries and packages used: transformers, torch, pillow, requests

Theory:

Multimodal AI systems are built to simultaneously process and integrate information from multiple input types such as text, images, audio, and video under a single unified model.

Image captioning is a classic multimodal task where the system generates or selects a textual description for a given image. Earlier methods employed CNNs to extract image

features and RNNs to generate corresponding text, whereas contemporary systems predominantly rely on transformer-based architectures for both tasks.

CLIP (Contrastive Language–Image Pretraining) is a robust multimodal model from OpenAI, trained via contrastive learning on an extensive collection of image-text pairs. Rather than producing captions token by token, CLIP builds a joint embedding space where semantically aligned images and texts are represented in close proximity, while mismatched pairs are pushed apart. This design enables the model to score similarity between an image and any set of candidate descriptions.

Working of CLIP for Caption Selection:

The model converts the input image into an image embedding and each candidate caption into corresponding text embeddings. Similarity scores are then calculated between the image embedding and all text embeddings, which are subsequently normalized into probabilities via the softmax function. The caption achieving the highest probability score is returned as the model's predicted description.

Methodology:

1. Install required libraries including transformers, torch, pillow, and requests.
2. Import CLIPProcessor and CLIPModel from the transformers library.
3. Load the pre-trained CLIP model “openai/clip-vit-base-patch32”.
4. Load an image from a URL using the requests library and PIL.
5. Define a list of possible captions describing different scenarios.
6. Preprocess both the image and text captions using CLIPProcessor.
7. Pass the processed inputs to the CLIP model to obtain similarity scores.
8. Apply softmax to convert similarity scores into probabilities.
9. Select the caption with the highest probability as the final prediction.
10. Display the predicted caption.

Use Cases:

1. Automated Image Annotation: Systems can automatically tag images with descriptive captions for large image databases.
2. Accessibility Tools: Helps visually impaired users by describing images in words.
3. Content Moderation: Detects and describes image content for filtering inappropriate material.
4. Visual Search Engines: Enables searching images using natural language queries.
5. Social Media Automation: Generates captions for uploaded images.

Advantages:

1. No need for task-specific training since CLIP is pre-trained on massive datasets.
2. Handles both image and text inputs effectively.
3. Flexible — works with any set of candidate captions.
4. High accuracy for general image understanding tasks.
5. Eliminates the need for complex caption-generation architectures.

Limitations:

1. Does not truly generate captions — only selects from given candidates.
2. Performance depends on the quality of provided captions.
3. May fail on highly specialized or domain-specific images.
4. Requires significant computational resources for large-scale use.
5. Cannot describe details not present in candidate captions.

Applications:

1. Image search and retrieval systems.
2. AI-powered photo management software.
3. Robotics vision systems.
4. Medical image annotation assistance.

5. E-commerce product image tagging.

Conclusion:

This assignment built a multimodal image captioning pipeline leveraging the CLIP model. The system initializes a pre-trained CLIP model, processes a given image alongside several candidate captions, and identifies the best-matching caption through similarity score comparison. The experiment highlights how multimodal architectures effectively connect visual and linguistic representations. While the system selects rather than generates captions, it offers an efficient and capable solution for image understanding built on state-of-the-art pre-trained foundations.